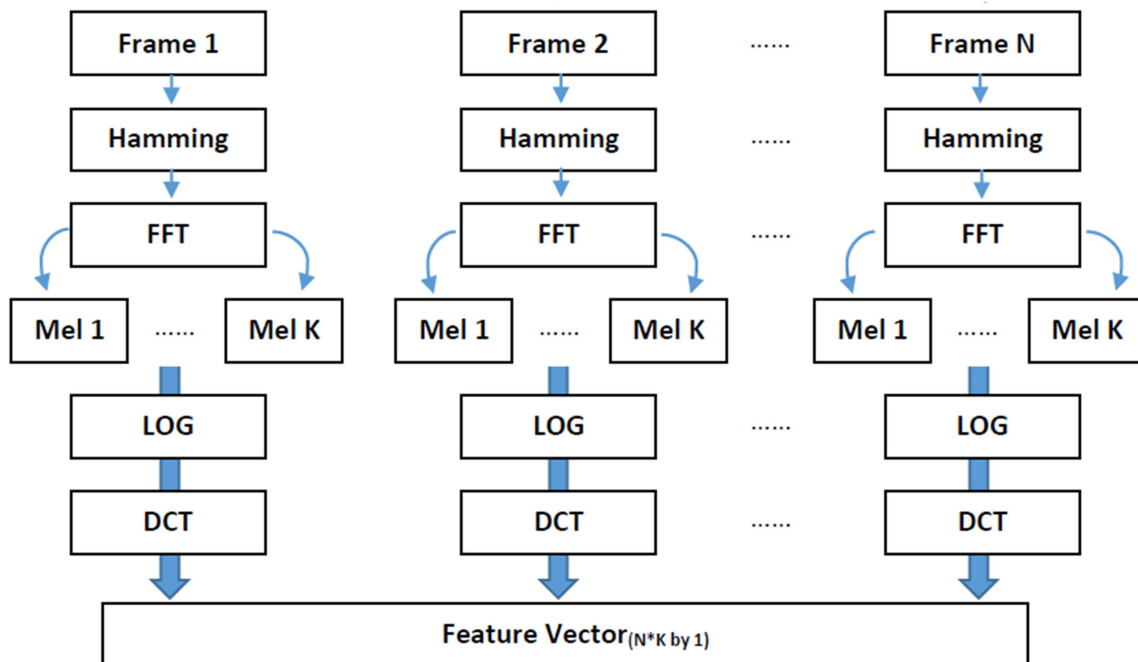


EEE 491 Lab Assignment

Lab-MATLAB: Modeling and Analysis of Spoken Number Recognition System

1. Technical Specifications

A model of a spoken number recognition system shall be constructed by using the Mel-Frequency Cepstrum in the MATLAB. The MATLAB program shall be interfaced with the microphone of your PC and spoken numbers shall be captured and stored in the MATLAB. Each spoken number shall be chopped in to about 30msec duration audio frames with about 50% overlapped (15msec) with the previous audio frame. You should research on the amount of frame durations and overlapping durations. Then for each frame, Mel-Frequency Cepstral Coefficients (MFCC) shall be calculated and gathered to have a feature vector for the spoken number. The MATLAB simulation shall be run for the numbers 0 to 9 in order to obtain feature vector for each number. The basic flow shall be as shown in the figure below. During your study, you may change the windowing type, number of Mel filter banks ($K > 31$), and DCT length (> 7).



The feature vectors shall be used for the recognition of a spoken number. After obtaining a feature vector of a spoken number, the new feature vector shall be compared with the pre-stored ten feature vectors. The recognition process shall be completed by identifying the number with the minimum Euclidean distance between the pre-stored feature vector and the new obtained feature vector. The recognized number shall be printed in the MATLAB window.

Keep in mind that your system that is designed and simulated on the MATLAB will be implemented on the FPGA (Basys-3 board). Therefore, an audio signal captured by the MATLAB must be quantized to 12-bits due to the 12-bit ADC module. You should also consider the resource limitations on the FPGA, so that you need to quantize the results at the output of each function as specified in the lab assignments.

2. Simulations and Test Results

Run MATLAB and observe outputs of each function on various plots. You should conclude on the correctness of each function output (that is the verification). You must emphasize the truncations that you did at each function output

data. You must also discuss the selected values for various parameters that are mentioned above. You must also tabulate the Euclidean distances between the feature vector of the spoken number and each feature vector of the numbers that were extracted before. Verify that the number is recognized with some probability.

3. Demonstration

- Speak to the microphone of the PC while running the MATLAB, and print the spoken number in the MATLAB window.
- You may be asked to run the functions step by step on the MATLAB.
- At least show the following during demonstration:
- Show that at each step the calculated value is quantized to the specified number of bits.
- Play the captured sound. Play the 12-bit quantized captured sound.
- Plot the samples of sound and show that start of speech is identified.
- Plot a frame of the speech and then plot its windowed frame.
- Plot Mel filters on the same plot as 'o' dots (not as connected lines) (Magnitude versus frequency (Hz) up to 4Khz)
- Plot FFT of a frame (Magnitude and also Log of Magnitude versus frequency up to 4KHz)
- Plot Energy and also Log of Energy versus frequency up to 4KHz for a frame.
- Plot DCT results for a frame.
- Print the recognized spoken number. Print or plot its distances to the other compared feature vectors.